

Project Title: DinnersReady

Project Overview

DinnersReady is an app that suggests recipes to the user based on current kitchen ingredient inventory. The aim is to help reduce food waste by tracking grocery inventory and to eventually include expiration data on fresh food items to help the user avoid tossing out a bunch of produce or meat at the end of the week.

Version 1 Implementation

- Inventory system: Kitchen inventory is created, updated, and saved to a local CSV file using pandas.
- Inventory menu interaction (CLI): Users can view inventory, as well as add or subtract ingredients and their quantities.
- Recipe search integration: Using TheMealDB's premium API, users can search for recipes that include multiple ingredients from their inventory.
- Recipe instructions lookup: Users can choose a specific recipe from a list of recipes populated by the multi-ingredient search. When a recipe is chosen, the user will be presented with detailed cooking instructions.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\cljon\OneDrive\Documents\GitHub\caitlinz_Dir
ljon\OneDrive\Documents\GitHub\caitlinz_DinnersReady\Dir
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==== Welcome to DinnersReady V1 ====
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==== Main Menu ====
1. Manage Inventory (View or Add Ingredients)
2. Search Recipes by Ingredients
3. Exit
Select an option (1-3): 1

==== Inventory Management ====
1. View Current Inventory
2. Add or Update Ingredient
3. Reduce or Remove Ingredient
4. Back to Main Menu
Select an option (1-4): 1

Your current ingredients:
ingredient quantity
chicken breast 2
tomato 4
garlic 1
onion 11
spinach 6
beef 9
cumin 10
basil 6
```

Figure 2, Viewing Current Inventory

```
==== Inventory Management ====
1. View Current Inventory
2. Add or Update Ingredient
3. Reduce or Remove Ingredient
4. Back to Main Menu
Select an option (1-4): 2
Enter ingredient: feta

Enter quantity for feta: 6
Added feta to inventory.

==== Inventory Management ====
1. View Current Inventory
2. Add or Update Ingredient
3. Reduce or Remove Ingredient
4. Back to Main Menu
Select an option (1-4): 1

Your current ingredients:
ingredient quantity
chicken breast 2
tomato 4
garlic 1
onion 11
spinach 6
beef 9
cumin 10
basil 6
feta 6
```

Figure 1.2, Adding Item and Quantity

```
==== Main Menu ====
1. Manage Inventory (View or Add Ingredients)
2. Search Recipes by Ingredients
3. Exit
Select an option (1-3): 2

Which ingredients would you like to use?
Enter ingredient(s) separated by commas: garlic, tomato
Searching for recipes with: ['garlic', 'tomato']...

We found 19 suggested recipes based on your ingredients.

Suggested Recipes:
1. Algerian Bouzgene Berber Bread with Roasted Pepper Sauce
2. Balchi di Pisca
3. Beef Empanadas
4. BEEF LOK LAK (Lok Lak Sach Ko)
5. Beef Mandi
6. Callaloo Jamaican Style
7. Chivito sandwich
8. Ezme
9. Falafel Pita Sandwich with Tahini Sauce
10. Imam bayildi with BBQ lamb & tzatziki
11. Mutabbaq
12. Pa Amb Tomaquet (Bread with Tomato)
13. Paella
14. Pastel de Papas (Chilean Potato Pie)
15. Spanish meatballs with clams, chorizo & squid
16. Spanish tomato bread with jamón Serrano
17. Vegetarian Casserole
18. Venezuelan Coconut Chicken
19. Venezuelan Shredded Beef
Select a recipe number (1-19) to view steps: 4
Cooking instructions for: BEEF LOK LAK (Lok Lak Sach Ko)
Fry the eggs sunny-side up and set aside. Finely chop the garlic. Wash the lettuce leaves and drain well. Thinly slice the onion and tomatoes.
Cut the beef tenderloin into 2-3 cm cubes. Marinate with sea salt, palm sugar, Chinese cooking wine, oyster sauce, soy sauce, and sesame oil. Mix
well.
Heat a frying pan on high heat with cooking oil. Add chopped garlic and stir until the garlic browns lightly. Add the marinated beef (with all th
e marinade) and cook for 5-7 minutes, depending on how rare you want the beef.
Dry roast the black Kampot peppercorns in a frying pan over medium-low heat for 2-3 minutes until fragrant. Grind the roasted pepper, place it in
a bowl, then add sea salt and lime juice. Mix well.
On a serving plate, arrange a bed of lettuce, then layer tomatoes, then onions. Spoon the cooked beef over the vegetables and place fried eggs on
top. Serve with the Kampot pepper sauce and fragrant steamed rice.
Sear hot and fast, Lok Lak is at its best when the beef stays tender and juicy, with a bold, bright dipping sauce to cut through the richness.
```

Figure 3, Multi-Ingredient Search and Recipe Selection Details

Obstacles Encountered

- I encountered a caching issue with VSCode that led me to implement and remove code multiple times until I was able to figure out the cause of the issues I was encountering. Restarting the program periodically after a commit is required.
- Blank spaces and extra spaces were causing issues when asking for user input to search for ingredients. Using `'[item.strip() for item in user_raw_split(",") if item.strip()]'` helped with this blank space / empty string issue.
- I was unable to search by multiple ingredients using the free version of TheMealDB API, but purchasing the premium version has been well worth it to get access to this functionality.

- I would like to implement a fallback method of utilizing the free version of TheMealDB API in case of issues with the premium version, as well as just searching one ingredient without having to use my unique API key. I can spend more time with this.

Where Help May Be Needed

- If I were to try to utilize the Edamam API alongside TheMealDB API to get estimated shelf-lives for ingredients, I would need your guidance in how to best set up these functions to run almost simultaneously. The shelf-life variables in Edamam would influence the prioritization of ingredient searches within TheMealDB.
- I have not used Streamlit before, but there seems to be plenty of helpful information and documentation on getting started, so I can probably figure out the basics. If there is more complex functionality that I'd need help with, I will reach out.

Upcoming Milestones

1. TheMealDB free API fallback logic (CLI): Allow the program to run with a public key ("1") provided in the free tier of TheMealDB API, if the premium version is not available. This would allow most functionality of the current version, but with only a solo ingredient search.
2. Streamlit GUI: Build a Streamlit front-end with clean display of inventory, as well as a click-to-modify (add, subtract, remove) table of ingredients.
3. Expiration tracking System: Add a 'purchase date' and 'expiration date' to inventory.csv, use Streamlit to save calendar tags.
4. ****Nice to have**** Edamam automates shelf-life: If purchase dates and expiration dates are not input by the user, tap Edamam to fill in a typical expiration window starting from the initial input day.

Self Reflection

I am satisfied with my progress so far. I have the basic functionality there, and feel like I can now take time to research ways I can build upon the basic functionality to make it better and more intuitive for a user that has no prior experience with my app. I am looking forward to learning how to use Streamlit, what I've been researching and reading about it makes me excited to really dig into the implementation. I do think that I'll be able to complete this project in the next few weeks, and if I get ahead I can also try to build in the 'nice to haves' that I've laid out.

Some of the functions and menu building came easier than expected, like calling TheMealDB API and returning the data in an easy to read way. There were a couple of times where things like adding a subtract function and trying to add a category search threw me for a loop when I least expected it, and I spent far more time than I expected trying to solve an issue with the information not being returned from the API

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only to discover that VSCode had a caching issue that wasn't allowing me to run and test my file properly after a few commits.

Overall I've really enjoyed this project so far, and feel more comfortable reading, trying, and understanding more of what others share on reddit or other developer forums when they're working through issues.